Engineering Chemistry

UNIT – 2 : Modern Engineering Materials

A clear, exam-ready tour of the materials behind modern technology — semiconductors, superconductors, supercapacitors and the carbon nanoworld — with hand-drawn diagrams.

Topics in this unit

1. Semiconductors — concept & applications
2. Superconductors — concept & applications
3. Supercapacitors — concept, classification & applications
4. Nanomaterials — classification & properties
5. Fullerenes, carbon nanotubes & graphene

> * Study tip
> Every topic has a small diagram. Learn the red words first, then redraw the diagram once from memory — that is the fastest way to revise for exams.

1. Modern Engineering Materials — The Big Picture

Behind every new technology — a faster phone chip, an MRI scanner, an electric-car battery, a lighter aircraft — there is a special material making it possible. This unit introduces five families of modern materials and what makes each one remarkable.

NOTE

Our journey: semiconductors (the heart of all electronics) → superconductors (zero resistance) → supercapacitors (super-fast energy storage) → nanomaterials (matter at the billionth-of-a-metre scale) → the carbon wonder-trio fullerenes, carbon nanotubes & graphene. One idea links them all: how freely electrons can move, set by a material's energy bands — so that is where we begin.

2. Semiconductors & Band Theory

A semiconductor is a material whose ability to conduct electricity lies between a conductor and an insulator. Silicon (Si) and germanium (Ge) are the famous examples — they run every computer chip in the world.

DEFINITION

Semiconductor: a solid whose electrical conductivity is between that of a conductor and an insulator, controlled by a small energy band gap and strongly affected by temperature and doping.

Band theory of solids

In a solid, millions of atomic energy levels merge into continuous energy bands. Two of them matter most:

- Valence band (VB) — the lower band, normally full of electrons.
- Conduction band (CB) — the upper band, normally empty; electrons here are free to move and carry current.
- Band gap (E_g) — the forbidden energy region between VB and CB. Its width decides how the solid behaves.

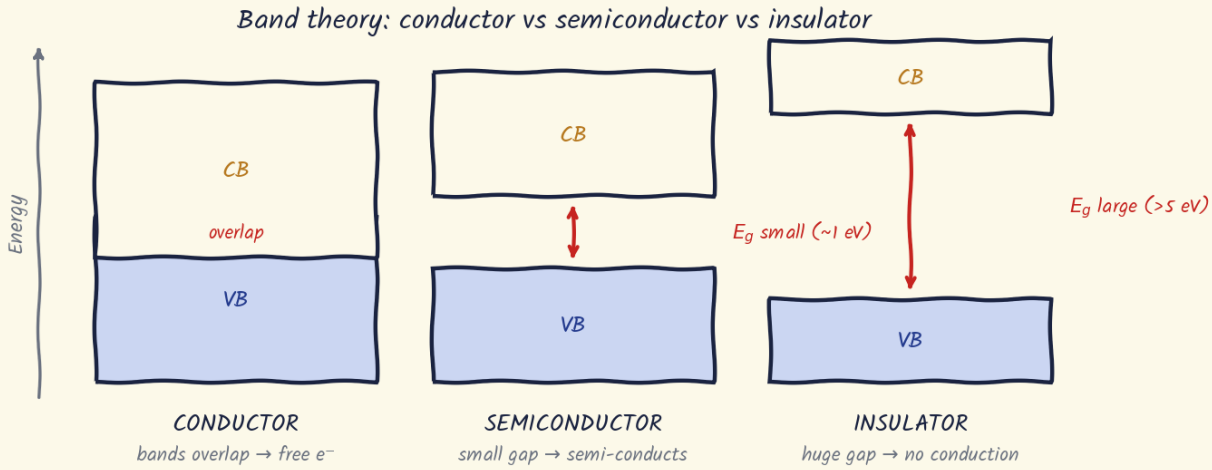


Fig 1 — Conductors have overlapping bands (no gap), semiconductors a small gap, insulators a large gap.

Conductor vs semiconductor vs insulator

Type	Band gap E_g	Example	Conducts?
Conductor	0 (bands overlap)	Cu, Al, metals	yes, freely
Semiconductor	small (~0.5–3 eV)	Si (1.1 eV), Ge (0.7 eV)	a little; more when hot
Insulator	large (>5 eV)	diamond	no

KEY POINT

Unlike metals, a semiconductor's conductivity increases as temperature rises. Heat gives more electrons enough energy to jump the small gap into the conduction band, so more charge carriers appear. (At 0 K a pure semiconductor behaves like an insulator — the idea behind thermistors.)

3. Intrinsic & Extrinsic Semiconductors

Semiconductors come in two kinds: pure ones and deliberately impure ones. Adding a tiny, controlled amount of impurity is the trick that makes all modern electronics possible.

Intrinsic (pure) semiconductors

DEFINITION

Intrinsic semiconductor: a pure semiconductor (just Si or Ge) with no added impurity. Its few carriers come only from electrons thermally excited across the gap, leaving behind an equal number of holes.

- Equal numbers of electrons (in CB) and holes (in VB).
- Very low conductivity at room temperature.

Extrinsic (doped) semiconductors

DEFINITION

Doping: adding a tiny amount of impurity (about 1 atom in a million) to a pure semiconductor to greatly raise its conductivity. The product is an extrinsic semiconductor.

n-type — doped with group 15

Add a pentavalent impurity (P, As — group 15). Four of its five valence electrons form bonds; the fifth is left over and almost free.

- The impurity is a donor — it sits as a donor level just below the conduction band and easily gives up its spare electron.
- Majority carriers = electrons (negative) → hence n-type.

p-type — doped with group 13

Add a trivalent impurity (B, Al — group 13). With only three valence electrons, one bond is left incomplete — a hole.

- The impurity is an acceptor — it sits as an acceptor level just above the valence band and readily accepts an electron.
- Majority carriers = holes (positive) → hence p-type.

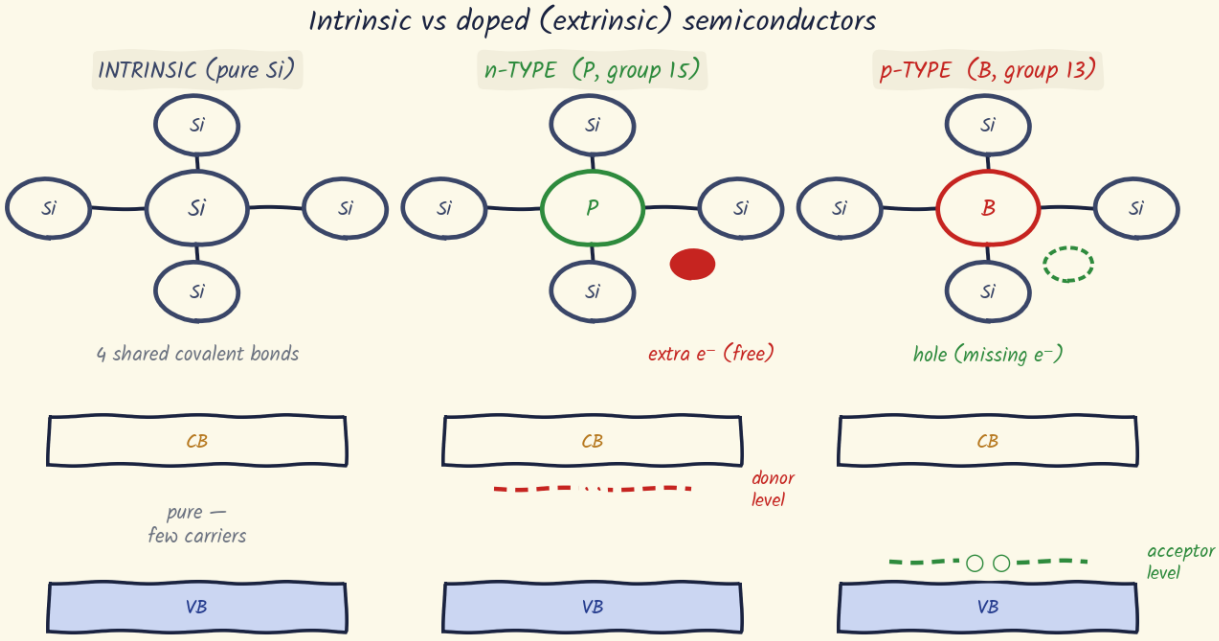


Fig 2 — Pure Si vs n-type (P donor, spare electron) vs p-type (B acceptor, hole), showing the donor and acceptor energy levels.

Feature	n-type	p-type
Dopant	group 15 (P, As) — pentavalent	group 13 (B, Al) — trivalent

Feature	n-type	p-type
Extra species	spare electron	hole
Majority carrier	electrons (−)	holes (+)
Impurity acts as	donor (level below CB)	acceptor (level above VB)

EXAMPLE

Joining a p-type piece to an n-type piece makes a p–n junction — the basis of the diode, the transistor and the solar cell, which are the building blocks of every chip.

Applications of semiconductors

- Diodes & rectifiers — convert AC to DC.
- Transistors & ICs (microchips) — switching and amplification in all computers.
- Solar (photovoltaic) cells — turn sunlight into electricity.
- LEDs, photodiodes & lasers — emitting and sensing light.
- Sensors — for temperature, light, pressure and gas.

4. Superconductors

Imagine a wire that carries current forever with no energy loss. That is a superconductor — one of the most striking effects in all of physics.

DEFINITION

Superconductor: a material that shows exactly zero electrical resistance when cooled below a certain critical temperature (T_c).

The effect was discovered by Kamerlingh Onnes in 1911, who cooled mercury and watched its resistance vanish at $T_c = 4.2$ K.

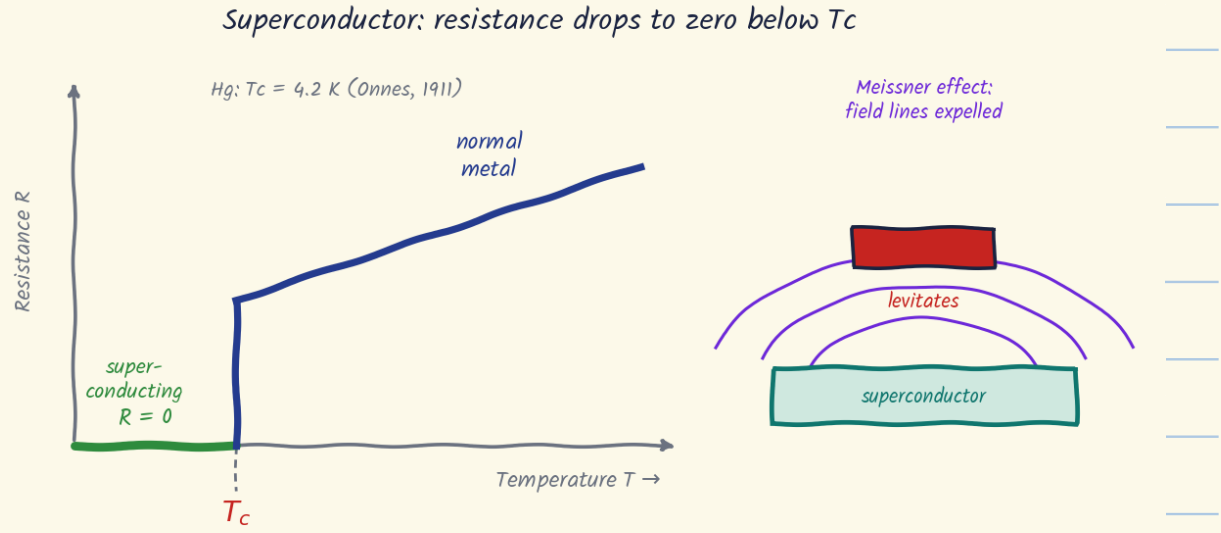


Fig 3 — Resistance drops sharply to zero at T_c; below T_c a magnet floats above the superconductor (Meissner effect).

Meissner effect

DEFINITION

Meissner effect: below T_c a superconductor completely expels magnetic field lines from its interior (perfect diamagnetism). This is why a magnet levitates above it.

KEY POINT

The Meissner effect is the true test of superconductivity and is the basis of magnetic levitation (Maglev) trains.

Type I and Type II

Feature	Type I (soft)	Type II (hard)
Made of	pure metals	alloys & ceramics

Feature	Type I (soft)	Type II (hard)
T_c	very low	higher
Magnetic field	expels fully, then breaks down	allows partial entry; stays super to high fields
Main use	little practical use	powerful magnets (MRI, Maglev)

A famous high-T_c ceramic is $YBa_2Cu_3O_7$ (YBCO), which superconducts at $T_c \approx 92$ K — above the boiling point of cheap liquid nitrogen (77 K).

BCS theory — why it happens

DEFINITION

BCS theory: below T_c the electrons pair up into **Cooper pairs** that move through the crystal lattice together **without scattering** — so there is no resistance.

Applications

- **MRI scanners** — superconducting magnets create the strong magnetic field.
- **Maglev trains** — frictionless magnetic levitation.
- **SQUIDs** — ultra-sensitive magnetometers.
- **Lossless power transmission** and very powerful electromagnets.
- **Particle accelerators** (e.g. the LHC).

5. Supercapacitors

A **supercapacitor** (or ultracapacitor) stores far more energy than an ordinary capacitor and charges far faster than a battery — it **bridges**

the gap between the two.

DEFINITION

Supercapacitor: an electrochemical energy-storage device that stores charge mostly physically (electrostatically) on very high-surface-area electrodes, giving very fast charge-discharge and a very long life.

Key properties

- Very fast charge and discharge (seconds).
- Very long cycle life — over 100,000 cycles.
- High power density, but lower energy density than a battery.

Property	Capacitor	Supercapacitor	Battery
Charge storage	electrostatic	mostly electrostatic	chemical
Charge speed	very fast	fast (seconds)	slow
Energy stored	very low	medium	high
Cycle life	huge	very long ($>10^5$)	limited

Classification

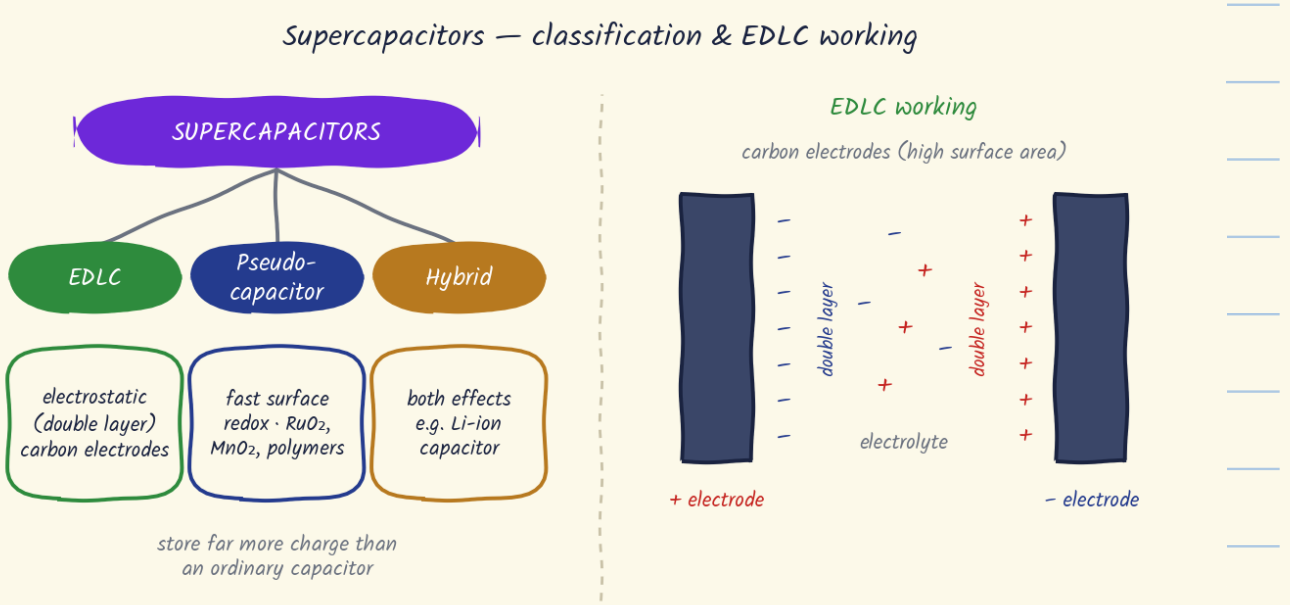


Fig 4 — The three types of supercapacitor and how an EDLC stores charge in the electric double layer.

1. EDLC — Electric Double-Layer Capacitor

Charge is stored electrostatically in the double layer that forms at the interface between a high-surface-area carbon electrode and the electrolyte. There is no chemical reaction — it is purely physical.

2. Pseudocapacitor

Stores charge through fast, reversible surface redox reactions (faradaic). It uses metal oxides (RuO_2, MnO_2) or conducting polymers.

3. Hybrid

Combines both mechanisms (electrostatic + redox) to store more energy. A common example is the lithium-ion capacitor.

Applications

- Regenerative braking in electric vehicles and trains.
- Memory and backup power and UPS systems.
- Camera flash and quick burst-power.

- Hybrid vehicles and buffering renewable energy (solar, wind).

6. Nanomaterials

A nanometre (nm) is one-billionth of a metre — about the width of a few atoms. Materials built at this scale behave in surprising new ways.

> DEFINITION
> Nanomaterial: a material with at least one dimension in the 1–100 nm range. At this size its properties differ greatly from the same material in bulk.

Why are nanomaterials special?

- Huge surface-area-to-volume ratio — most atoms sit on the surface, so the material is far more reactive and makes an excellent catalyst.
- Quantum confinement — electrons squeezed into tiny spaces change the optical, electrical and magnetic behaviour (e.g. colour can depend on particle size).

Classification by dimension

Nanomaterials are grouped by how many of their dimensions lie outside the nanoscale — giving 0-D, 1-D, 2-D and 3-D types.

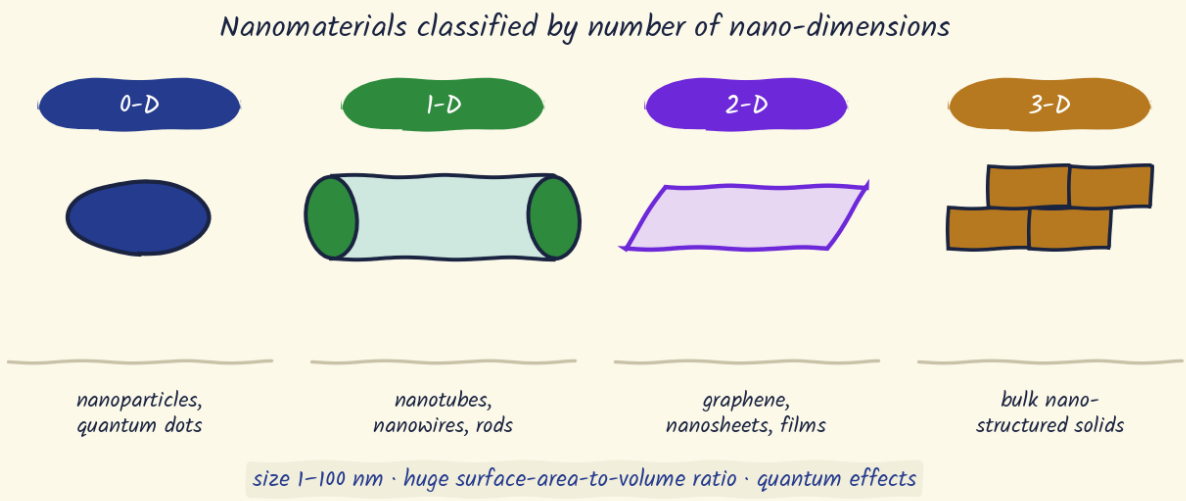


Fig 5 — Nanomaterials classified as 0-D, 1-D, 2-D and 3-D by the number of nano-sized dimensions.

- 0-D — every dimension nanoscale: nanoparticles, quantum dots.
- 1-D — one dimension larger: nanotubes, nanowires, nanorods.
- 2-D — two dimensions larger: graphene, nanosheets, thin films.
- 3-D — bulk nanostructured solids built from nanoscale grains.

KEY POINT

Same material, new powers: bulk gold is inert and yellow, but gold nanoparticles are brightly coloured and chemically active. At the nanoscale, size itself becomes a property.

7. Fullerenes — C_{60}

The carbon wonder-trio begins with the fullerene — a football-shaped molecule of pure carbon, discovered in 1985.

DEFINITION

Fullerene (C_{60}): a closed, hollow cage of 60 carbon atoms arranged as 20 hexagons + 12 pentagons — exactly like a soccer ball. Its full name is Buckminsterfullerene.

Structure & properties

- 60 sp^2 carbon atoms forming a spherical cage.
- Hollow and remarkably stable.
- A good electron acceptor (takes up electrons easily).

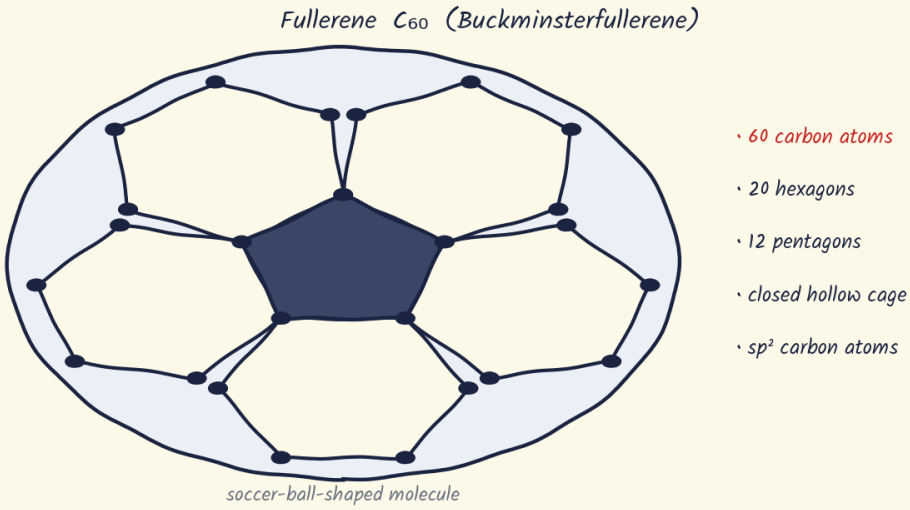


Fig 6 — C_{60}: a closed cage of 60 carbons — 20 hexagons and 12 pentagons — shaped like a soccer ball.

Applications

- Drug delivery — molecules carried inside the cage.
- Lubricants — act like tiny ball-bearings.
- Catalysts and organic solar cells.
- Superconductors (when doped) and antioxidants.

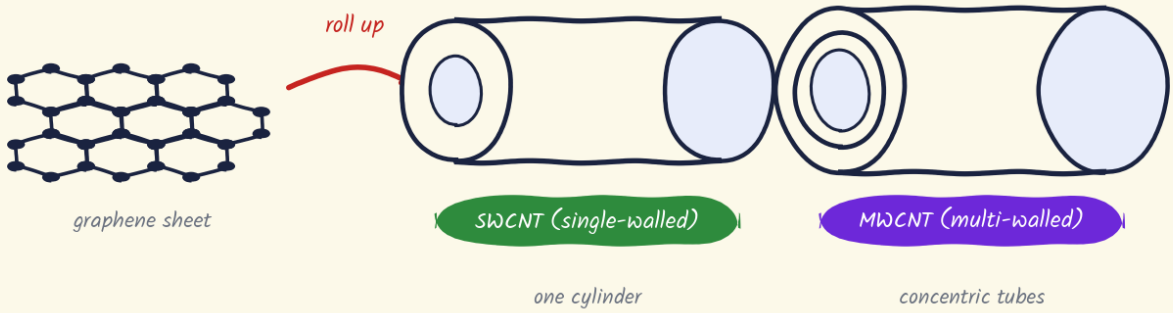
8. Carbon Nanotubes (CNTs)

Take a sheet of graphene and roll it into a cylinder — you get a carbon nanotube, one of the strongest materials ever made.

DEFINITION

Carbon nanotube (CNT): a rolled-up cylinder of graphene (sp^2 carbon). It may be **single-walled (SWCNT)** — one tube — or **multi-walled (MWCNT)** — several concentric tubes.

Carbon nanotube: a rolled-up graphene sheet



~100× stronger than steel, about one-sixth the weight · great conductor

Fig 7 — A graphene sheet rolls into a tube: SWCNT (one wall) and MWCNT (several concentric walls).

Properties

- About **100× the tensile strength of steel at only one-sixth the weight**.
- Excellent **electrical and thermal conductors**.
- Very high stiffness combined with flexibility.

Applications

- **Composites** — strong, light materials for sports gear and aerospace.
- **Nano-electronics** — transistors and field-emission displays.
- **Sensors** and **hydrogen storage**.
- **Electrodes** for batteries and supercapacitors.

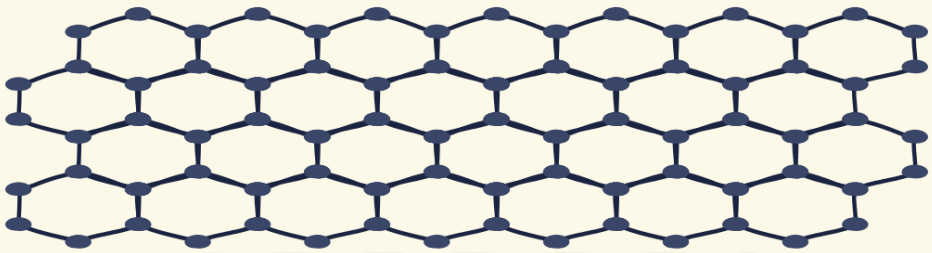
9. Graphene

Graphene is a single layer of carbon atoms — just one atom thick — yet it is the strongest material ever measured. Its discoverers won the 2010 Nobel Prize in Physics.

DEFINITION

Graphene: a single 2-D sheet of sp^2 carbon atoms arranged in a honeycomb (hexagonal) lattice — the basic building block of graphite, carbon nanotubes and fullerenes.

Graphene: a one-atom-thick honeycomb of sp^2 carbon



- each dot = C atom
- lines = C–C bonds
- 2-D sheet, 1 atom thick
- strongest known material

transparent · flexible · excellent heat & electrical conductor

Fig 8 — Graphene: a one-atom-thick honeycomb sheet of sp^2 carbon atoms.

Properties

- Strongest known material, yet light and flexible.
- Excellent electrical and thermal conductor.
- Transparent — lets light pass through.

Applications

- Fast transistors and flexible, transparent electronics.
- Sensors and battery and supercapacitor electrodes.

- Composites and desalination membranes.

> KEY POINT
> Graphene, CNTs and fullerenes are all made of the same sp^2 carbon — just shaped differently: a flat sheet, a rolled tube and a closed ball.

10. Applications & Importance

Together, these modern materials power nearly every advanced technology around us:

- Electronics & computing: semiconductor chips, transistors, LEDs and solar cells.
- Medicine: superconducting MRI magnets; fullerene drug delivery; CNT and graphene biosensors.
- Transport: Maglev trains, supercapacitor regenerative braking, and light CNT and graphene composites.
- Energy: supercapacitors and graphene electrodes for fast storage; photovoltaic cells; lossless power lines.
- Everyday materials: stronger, lighter sports and aerospace parts from nanocomposites.

QUICK RECAP

- Semiconductors: small band gap (Si 1.1 eV); conductivity rises with temperature and doping.
- n-type = group-15 doping (extra electrons, donor level); p-type = group-13 doping (holes, acceptor level).
- Superconductors: zero resistance below T_c, expel magnetic fields (Meissner effect); Cooper pairs (BCS).
- Supercapacitors: fast charge, very long life; types EDLC, pseudocapacitor, hybrid.
- Nanomaterials (1–100 nm): huge surface area + quantum effects; types 0-D, 1-D, 2-D, 3-D.
- Carbon trio: fullerene C_{60} (cage), CNT (rolled tube, very strong), graphene (one-atom sheet, strongest).

EXAM-ORIENTED QUESTIONS

Q1. What is a semiconductor? How does its conductivity change with temperature? [3 marks]

Ans. A semiconductor has conductivity between a conductor and an insulator, set by a small band gap (~0.5–3 eV). Unlike metals, its conductivity increases with temperature because heat excites more electrons across the gap into the conduction band.

Q2. Explain band theory for conductors, semiconductors and insulators. [5 marks]

Ans. Solids have a filled valence band and an empty conduction band separated by a band gap E_g. Conductors: bands overlap (no gap). Semiconductors: small gap (~1 eV). Insulators: large gap (>5 eV, e.g. diamond).

Q3. Distinguish between intrinsic and extrinsic semiconductors. [3 marks]

Ans. Intrinsic = pure Si or Ge with equal electrons and holes and low conductivity. Extrinsic = doped with a small impurity to raise conductivity, giving n-type or p-type.

Q4. Differentiate n-type and p-type semiconductors. [5 marks]

Ans. n-type: doped with group-15 (P, As); extra electrons are majority carriers; donor level lies just below the conduction band. p-type: doped with group-13 (B, Al); holes are majority carriers; acceptor level lies just above the valence band.

Q5. What is a superconductor and the critical temperature T_c? [3 marks]

Ans. A superconductor has exactly zero electrical resistance below a critical temperature T_c. It was discovered by Onnes (1911) in mercury at T_c = 4.2 K.

Q6. State the Meissner effect and give one application. [3 marks]

Ans. Below T_c a superconductor completely expels magnetic field lines (perfect diamagnetism), so a magnet levitates above it. It is the basis of Maglev trains and is used in MRI magnets.

Q7. Give the BCS idea and compare Type I and Type II superconductors. [5 marks]

Ans. BCS: below T_c electrons form Cooper pairs that move without resistance. Type I (soft, pure metals, very low T_c) expels the field fully; Type II (hard alloys and ceramics, higher T_c such as YBCO ≈ 92 K) is used in powerful magnets.

Q8. What is a supercapacitor? Name its three types. [3 marks]

Ans. An electrochemical device that stores charge mostly electrostatically on high-area electrodes — very fast charge-discharge and very long life, bridging a capacitor and a battery. Types: EDLC, pseudocapacitor and hybrid.

Q9. Define nanomaterials and give their classification by dimension. [5 marks]

Ans. Materials with at least one dimension in 1–100 nm, giving a huge surface area and quantum effects. Classified as 0-D (nanoparticles, quantum dots), 1-D (nanotubes, nanowires), 2-D (graphene, sheets) and 3-D (bulk nanostructured).

Q10. Describe the structure of fullerene C_{60} and give two uses. *[3 marks]*

Ans. C_{60} is a closed hollow cage of 60 sp^2 carbons (20 hexagons + 12 pentagons), shaped like a soccer ball. Uses: drug delivery and catalysts or organic solar cells.

Q11. What are carbon nanotubes? Compare SWCNT and MWCNT and give one property. *[3 marks]*

Ans. CNTs are rolled-up graphene cylinders. SWCNT has a single wall; MWCNT has several concentric walls. They are about 100× stronger than steel at one-sixth the weight and are excellent conductors.